Started on	Monday, 28 September 2026, 10:03 PM
State	Finished
Completed on	Monday, 28 September 2026, 10:15 PM
Time taken	12 mins 18 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Problem Statement:

Given a sorted array of integers say arr[] and a number x. Write a recursive program using divide and conquer strategy to check if there exist two elements in the array whose sum = x. If there exist such two elements then return the numbers, otherwise print as “No”.

Note: Write a Divide and Conquer Solution

Input Format

First Line Contains Integer n – Size of array

Next n lines Contains n numbers – Elements of an array

Last Line Contains Integer x – Sum Value

Output Format

First Line Contains Integer – Element1

Second Line Contains Integer – Element2 (Element 1 and Elements 2 together sums to value “x”)

Answer: (penalty regime: 0 %)

```c
#include<stdio.h>
int search(int a[],int l,int r,int x)
{
    if(l>r)
      return -1;
    int m=(l+r)/2;
    if(a[m]==x)
     return m;
    if(x<a[m])
     return search(a,l,m-1,x);
    return search(a,m+1,r,x);
}
int main()
{
    int n,x,i,p;
    scanf("%d",&n);
    int a[n];
    for(i=0;i<n;i++)
     scanf("%d",&a[i]);
    scanf("%d",&x);
    for(i=0;i<n;i++)
    {
        p=search(a,i+1,n-1,x-a[i]);
        if(p!=-1)
        {
            printf("%d\n",a[i]);
            printf("%d\n",a[p]);
            return 0;
        }
    }
    printf("No");
    return 0;
}
```

	Input	Expected	Got	
✔	4 2 4 8 10 14	4 10	4 10	✔

	Input	Expected	Got	
✔	5 2 4 6 8 10 100	No	No	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.